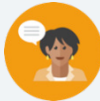


11.5 Interior and Exterior Angles of Triangles

Lesson Introduction

 Benchmark Focus	MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.		 Learning Targets	- I can show that the sum of the interior angles of a triangle is 180°. - I can show that the sum of the exterior angles of a triangle is 360°. - I can solve problems involving postulates, relationships, and theorems of triangles.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.		N/A	
 Essential Questions	- How can you show that the sum of interior angles of a triangle is 180°? - How can you show that the sum of the exterior angles of a triangle is 360°?			
 Lesson Narrative	Students discover how to show that the sum of the interior angles of a triangle is 180° , and that the sum of the exterior angles of a triangle is 360° . Students make, test, and revise conjectures based on relationships among angles that they observe in geometric constructions, including a circumscribed circle of a triangle, a circle inscribed in a triangle, and the sectors formed from the exterior angles of a triangle.			
 Component Summary	11.5.1 Warm-Up (~5 min.)	Share and justify superpowers of geometry superhero and villain (<i>SE p. 202</i>)	11.5.3 Activity (10-15 min.)	Discover relationships among exterior angles of triangles (<i>SE p. 204</i>)
	11.5.2 Exploration (15 min.)	Make conjectures about the relationship among interior angles of triangles (<i>SE p. 203</i>)	11.5.4 Guided Practice (10 min.)	Practice finding interior and exterior angles of triangles (<i>SE p. 206</i>)
	11.5.5 Wrap-Up (~5 Min)	Students summarize their learning (<i>SE p. 207</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Exterior Angle - Triangle Sum Theorem <u>Additional Glossary:</u> - Interior Angle		- Paper - Scissors - Straightedge - Compass (Optional) Colored Pencils	
	Additional Preparation			
	Prior to the lesson, work through 11.5.2 Exploration, 11.5.3 Activity, and 11.5.4 Guided Practice to ensure that you are familiar with the steps students will take and be prepared to support them as they work.			

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not understand that the sum of the interior angles of a triangle is 180°. - Students may not understand that the sum of the exterior angles of a triangle is 360°. - Students may understand the relationships but make errors in their calculations.	- Consider creating and displaying an Anchor Chart with the relationship between the interior and exterior angles in triangles for students to reference. - Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the relationship between the interior and exterior angles in triangles.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In 11.5.2 Exploration, prompt students to write down things they notice and wonder about the interior angles of the triangles from question 1 and question 2. This will allow students to talk through the relationship of the interior angles of triangles before writing a conjecture.	MA.K12.MTR.4.1: Discussion and Reflection During 11.5.2 Exploration and 11.5.3 Activity, students write a conjecture about the relationship among interior and exterior angles of a triangle. Students will have the opportunity to discuss their conjectures with their classmates. Encourage students to reflect on their initial conjectures and revise them as needed based on the conversations they have with their peers.
 Differentiate Instruction	Consider creating an anchor chart that displays all the relationships discovered throughout this unit, along with a specific example or two of each being applied as a reference point for students to use as they complete this lesson and the remainder of the unit.	
Lesson Notes:		

11.5.1 Warm-Up: Superhero

Component Overview: Students share and justify the superpowers of a geometry superhero and villain.



11.5.1 Warm-Up: Superhero

1. If geometry had a superpower, it would be ...

Answers will vary

because ...

Answers will vary

2. Geometry's villain would be ...

Answers will vary

because ...

Answers will vary



These prompts are intended to inspire student curiosity and interest. This component can help build students' agency and mathematical identities, as each student can participate. Consider having students draw and label a diagram of the superhero and villain.

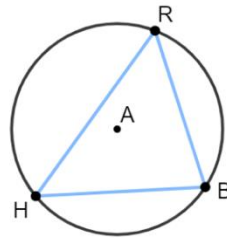
11.5.2 Exploration: Interior Angles of Triangles

Component Overview: Students discover how to show that the sum of the interior angles of a triangle is 180° . Students make, test, and revise conjectures based on relationships among angles that they observe in a circumscribed circle of a triangle and a circle inscribed in a triangle.



11.5.2 Exploration: Interior Angles of Triangles

1. Circle A is shown, where $m\widehat{RB} = 102^\circ$, $m\widehat{BH} = 108^\circ$, and $m\widehat{HR} = 150^\circ$.

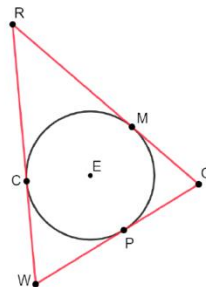


- a. Complete the table.

Angle Measures	
$m\angle RHB =$	51°
$m\angle RBH =$	75°
$m\angle HRB =$	54°

- b. What is the sum of the angles in $\triangle RBH$?
 180°

2. Circle E is shown where points P , C , and M are on the circle. $\overline{RG}$ and $\overline{RW}$ are tangent to Circle E and intersect at point R outside the circle. $\overline{GR}$ and $\overline{GW}$ are tangent to Circle E and intersect at point G outside the circle. $\overline{WR}$ and $\overline{WG}$ are tangent to Circle E and intersect at point W outside the circle. $m\widehat{CM} = 136^\circ$, $m\widehat{MP} = 108^\circ$, and $m\widehat{CP} = 116^\circ$.



- a. Complete the table.

Angle Measures	
$m\angle WRG =$	44°
$m\angle RGW =$	72°
$m\angle GWR =$	64°



This activity is an exploration of interior angles of triangles. Be careful not to give away too much information when supporting students during this activity. This activity has a natural unfolding process through discovery and exploration.

11.5.2 Exploration: Interior Angles of Triangles (continued)

- b. What is the sum of the angles in $\triangle RGW$?
 180°
- c. Write a conjecture about the sum of the interior angles of a triangle.
The sum of the interior angles of a triangle is 180° .
- d. Ask two classmates for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials
<i>Answers will vary</i>		

- e. Review your conjecture with your partner. If necessary, make any changes.



Triangle Sum Theorem

The sum of the measures of the interior angles of a triangle is 180° .



Notice and Wonder

Prompt students to write down things they notice and wonder about the sum of the interior angles of the triangles from question 1 and question 2. This will allow students to talk through the relationship of the interior angles of triangles before writing a conjecture.



Allowing students to share their conjecture with peers will provide entry points for auditory learners. Prompt students' by asking them if they agree or disagree with their peers' conjectures and why.

11.5.3 Activity: Exterior Angles of Triangles

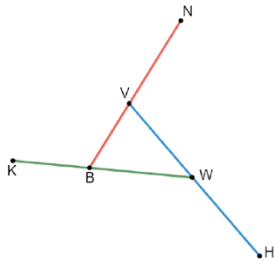
Component Overview: Students discover how to show that the sum of the exterior angles of a triangle is 360° . Students make, test, and revise conjectures based on relationships among angles that they observe in a geometric construction of the sectors formed from the exterior angles of a triangle.



11.5.3 Activity: Exterior Angles of Triangles

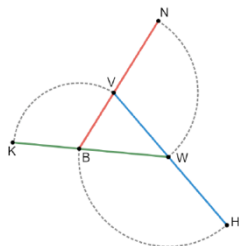
Work with your partner to complete the following.

- On a separate sheet of paper, draw three line segments, $\overline{BN}$, $\overline{VH}$, and $\overline{KW}$, to make $\triangle VWB$ as shown.



An **exterior angle** is an angle created by extending one leg of a triangle.

- List the exterior angles of $\triangle VWB$.
 $\angle VBK, \angle NVW, \angle BWH$
 - Estimate the sum of the exterior angles of $\triangle VWB$.
Answers will vary. 360°
- Use a compass to draw $\widehat{BH}$ using point W as the center, $\widehat{KV}$ using point B as the center, and $\widehat{NW}$ using point V as the center, as shown.



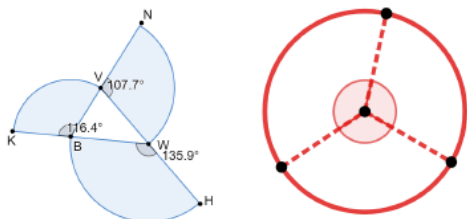
For this activity, provide students with a piece of paper, scissors, and a straightedge. Students will work through the scaffolded questions with a partner to discover that the sum of the exterior angles of a triangle is 360° . Consider reviewing this activity prior to the lesson to assist struggling students.



Consider providing struggling students access to a video that explains the steps of discovering that the sum of the exterior angles of a triangle is 360° . This will allow students the opportunity to pause and rewind the video as needed.

11.5.3 Activity: Exterior Angles of Triangles (continued)

- a. Trace the figure above onto patty paper and cut out each sector. Rearrange the sectors to determine the sum of the exterior angles.



- b. What do you notice about the sum of the exterior angles of the triangle?
They add to be 360°.
- c. Write a conjecture about the sum of the exterior angles of a triangle.
The sum of exterior angles of a triangle is 360°.
- d. Ask two classmates for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials
<i>Answers will vary</i>		
<i>Answers will vary</i>		

- e. Review your conjecture with your partner. If necessary, make any changes.



Consider using a dynamic tool, such as GeoGebra, to enable visual and kinesthetic students to investigate the relationship between the interior and exterior angles in triangles.



Consider challenging students who finish this activity early by having students verify if their conjecture will always be true. Students can create three line segments that form a triangle and complete the steps in question 2 to confirm if their conjecture is still true.



Listen as students discuss their thinking in response to their conjecture to determine student thinking to highlight, and/or misconceptions to address.

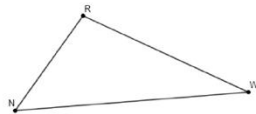
11.5.4 Guided Practice: Interior and Exterior Angles of Triangles

Component Overview: Students practice finding interior and exterior angles of triangles. Encourage students to show all of their work and check their initial calculations. During the debrief discussion, call on a variety of students to share their strategies and answers. Remind students to include the degree symbol with their answers.



11.5.4 Guided Practice: Interior and Exterior Angles of Triangles

1. $\triangle RWN$ is shown where $m\angle NRW = 14x + 3$, $m\angle RWN = \frac{9}{2}x - \frac{5}{2}$, and $m\angle WNR = 9x - 13$.

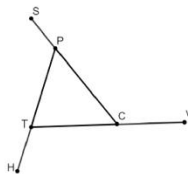


- a. Find the value of x .
 $x = 7$

- b. Complete the table.

$m\angle NRW = 101^\circ$	$m\angle RWN = 29^\circ$	$m\angle WNR = 50^\circ$
---------------------------	--------------------------	--------------------------

2. $\triangle PCT$ is shown, where $\angle SPT$, $\angle PCV$, and $\angle HTC$ are exterior angles. $m\angle PCT = 52^\circ$, $m\angle SPT = (11x - 20)^\circ$, and $m\angle HTC = (8x + 5)^\circ$.



- a. Find the value of x .
 $x = 13$
- b. What is the measure of $\angle SPT$?
 123°
- c. What is the measure of $\angle HTC$?
 109°



Common Misconceptions

Students may understand the relationships of interior and exterior angles of triangles, but still result in an incorrect answer. Consider asking students to explain their thought process when solving to determine if an incorrect answer is a result of students struggling with the new material or a miscalculation in solving.



- How did you find the answer?
- Why did you choose this strategy?
- How can you check that your answers are correct?

11.5.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



11.5.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.